

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, C, Java, JavaScript, TypeScript, SQL, Bash

Backend & APIs

Node.js, Express, FastAPI, REST APIs, async services & queues, multi-tenant systems

Testing & Tooling

pytest, unit & integration testing, debugging, code review, Agile / Scrum, Git, GitHub Actions, CI/CD

Systems & Performance

Apache Spark / PySpark, Spark UI, Linux profiling, performance optimization, evaluation metrics & benchmarking, PyTorch, NumPy

Cloud & DevOps

AWS, GCP, Azure, Docker, Linux

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered multi-tenant production backend systems (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; monitored production logs, **debugged and fixed** incidents, and added **integration tests** to protect against regressions.
- Built a full-stack data tooling platform (React, Node.js, MongoDB) used daily by the team, cutting manual work by **100+ hours/month** and improving throughput by **60%**, while working in an **Agile/Scrum** environment with sprint planning, stand-ups, and retrospectives.
- Reviewed and altered programs to adapt to new requirements and improve efficiency, routing tech-debt PRs and code migrations through a structured review workflow, and remediated **20+ VAPT findings** across the backend.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ REST APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability through targeted refactoring.

Projects

AgenticSearch - Staged Planning & Extraction Pipeline

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage system that **plans** typed retrieval, executes a staged pipeline (rerank, extraction, output verification), and returns cell-level provenance, hardened with **150+ automated tests** and pipeline instrumentation.
- Added an **evaluation harness** and provider routing / fallback to keep behavior predictable across backends; deployed as a live demo on Google Cloud Run.

AutoEval - Evaluation-Metric Design Agent

github.com/Deva-1903/AutoEval

Python, IR evaluation tooling, Git automation

- Built an agent that **designs and runs evaluation metrics**, proposing queries and labels, scoring against multiple systems, and accepting only updates that improve discriminative power on a holdout-heavy objective.
- Added guardrails (schema checks, duplicate rejection, read-only holdout protection) and Git-backed auditable results with report cards for a reproducible workflow.

Spark ETL Performance Optimization - Big-Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL | ~1.4B-row dataset

- Profiled and optimized a Spark pipeline over a **~1.4B-row** dataset to **increase operating efficiency**, cutting runtime **~25%** and tail-latency ratio (P99/P50) from **2.14x to 1.74x** via broadcast joins, Adaptive Query Execution, skew-join handling, and partitioning; analyzed stage-level metrics in **Spark UI**.
- Added a 6-rule **data-quality filter** (retaining 95.26% of records), schema-aware batched reads with column pruning, and filter pushdown, and documented a repeatable optimization playbook from stage-level analysis.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.